

Erdős Problem #460

EP-460

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $a_0 = 0$ and $a_1 = 1$, and in general define a_k to be the least integer $> a_{k-1}$ for which $(n - a_k, n - a_i) = 1$ for all $0 \leq i < k$. Does $\sum_{0 < a_i < n} \frac{1}{a_i} \rightarrow \infty$ as $n \rightarrow \infty$? What about if we restrict the sum to those i such that $n - a_j$ is divisible by some prime $\leq a_j$, or the complement of such i ?

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